

Foundation (Jalapeno)

- Q1.** What is the effect of the transformation $y = f(x) + 2$ on the graph of $y = f(x)$?
- (A) Shift down 2 units
(B) Shift up 2 units
(C) Shift left 2 units
(D) Shift right 2 units
(E) No change
- Q2.** Which equation represents a reflection of $y = f(x)$ over the x-axis?
- (A) $y = f(x)$
(B) $y = -f(x)$
(C) $y = f(-x)$
(D) $y = -f(-x)$
(E) $y = f(x) + 1$
- Q3.** What is the effect of the transformation $y = 2f(x)$ on the graph of $y = f(x)$?
- (A) Vertical compression by a factor of 2
(B) Vertical stretch by a factor of 2
(C) Horizontal compression by a factor of 2
(D) Horizontal stretch by a factor of 2
(E) No change
- Q4.** Which equation represents a horizontal shift of $y = f(x)$ by 3 units to the right?
- (A) $y = f(x - 3)$
(B) $y = f(x + 3)$
(C) $y = f(x) + 3$
(D) $y = f(x) - 3$
(E) $y = -f(x)$
- Q5.** What is the effect of the transformation $y = f(-x)$ on the graph of $y = f(x)$?
- (A) Reflection over the y-axis
(B) Reflection over the x-axis
(C) Horizontal shift by 1 unit to the right
(D) Vertical shift by 1 unit up
(E) No change
- Q6.** Which equation represents a vertical compression by a factor of 3 of $y = f(x)$?
- (A) $y = 3f(x)$
(B) $y = \frac{1}{3}f(x)$
(C) $y = f(3x)$
(D) $y = f(\frac{1}{3}x)$
(E) $y = f(x) + 1$
- Q7.** What is the effect of the transformation $y = f(x) - 1$ on the graph of $y = f(x)$?
- (A) Shift up 1 unit
(B) Shift down 1 unit
(C) Shift left 1 unit
(D) Shift right 1 unit
(E) No change
- Q8.** Which equation represents a reflection of $y = f(x)$ over the y-axis and a vertical shift up 2 units?
- (A) $y = f(-x) + 2$
(B) $y = -f(x) + 2$
(C) $y = f(x) - 2$
(D) $y = -f(-x) - 2$
(E) $y = f(x) + 2$

Open Ended Questions (FRQ)

- Q9.** Graph the function $y = 2f(x - 1) - 3$ if $f(x) = x^2$. Identify the parent function and describe the transformations applied.
- Q10.** Compare the graphs of $y = f(x)$ and $y = f(x + 2) - 1$. How do the graphs differ? What are the effects of the transformations on the graph of $y = f(x)$?

Chef's Tips

- Tip 1: Ensure students understand the parent function before applying transformations.
- Tip 2: Emphasize the effects of vertical and horizontal shifts, reflections, stretches, and compressions on the graph.
- Tip 3: Encourage students to sketch the graphs to visualize the transformations.